Difference-in-Differences

Beyond the Switch: Doses, Treatments that Turn On and Off

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Beyond the switch

- In chapters 4 and 5, the treatment is **binary and absorbing**: once a unit is treated, it remains treated.
- Two other cases are frequent:
  - a **dose**: the treatment has an intensity;
  - **on and off**: the treatment starts, stops, and may start again.
- Dose: **the China shock** (Autor et al. 2013). US local labour markets are exposed to Chinese imports according to their industry mix.
- On and off: **newspapers and democracy** (Gentzkow et al. 2011). According to models of voting under imperfect information, better-informed citizens vote more. Gentzkow et al. test this prediction on US counties that gain and lose daily newspapers between 1869 and 1928 (before radio and television, the newspaper was the main source of political information), and find that turnout rises with the number of newspapers.

References for the whole chapter: Baker, Callaway, Cunningham, Goodman-Bacon & Sant'Anna (2026), Appendix A; Callaway, Goodman-Bacon & Sant'Anna (2024); de Chaisemartin & D'Haultfœuille (2020, 2024).

Roadmap

1. **Doses.**
2. **Treatments that turn on and off.**
3. **Conclusion.**

Section 1

Doses

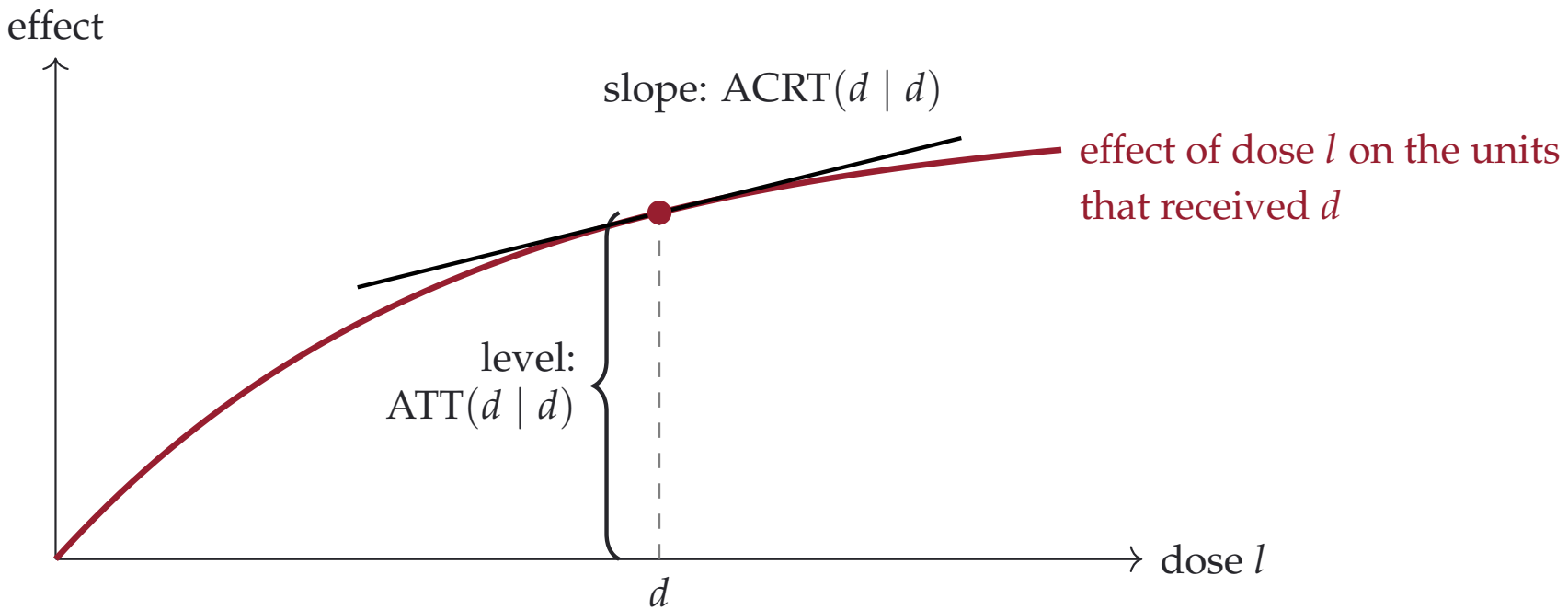
TWFE with a dose

$$y_{it} = f_i + f_t + \beta D_i \cdot \text{Post}_t + u_{it}$$

- The usual regression puts the dose in place of the treatment dummy: **one coefficient** for all doses.
- First problem: **some doses count more than their number of units.** $\hat{\beta}$ averages the effects of all doses with weights that differ from the share of units at each dose (Callaway et al. 2024).
- Second problem: $\hat{\beta}$ compares units with different doses. Such a comparison mixes two effects of a dose, which require different assumptions.

Two effects of a dose

- ATT$(d \mid d)$: the effect of the dose d on the units that received it.
- ACRT$(d \mid d)$: the effect of **one more unit** of dose on those units, the slope of the dose-response.
- China shock: the job loss of a highly exposed market, against the extra loss from a little more exposure.

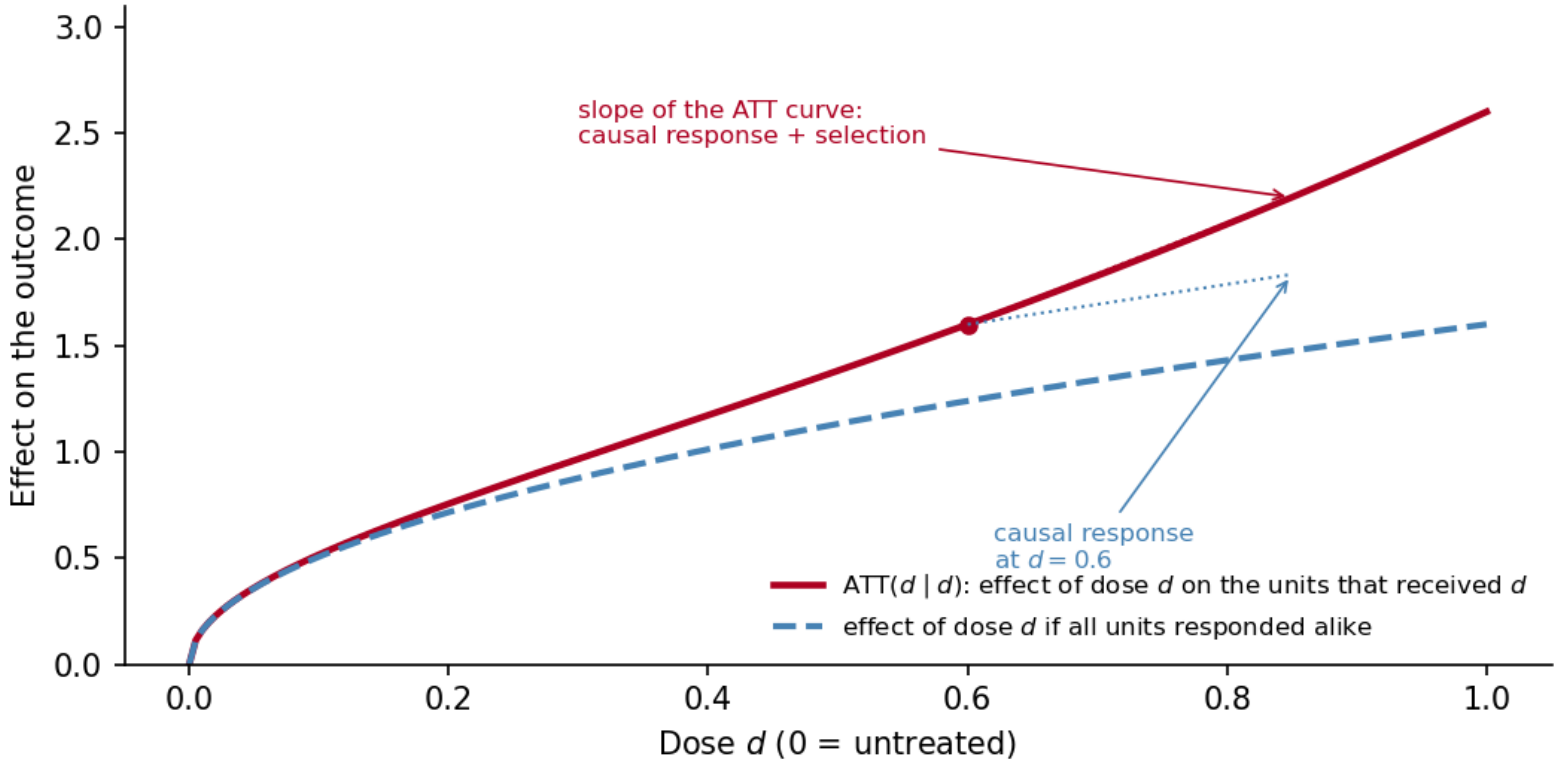


Two effects, two assumptions

Effect	Comparison	Assumption
Level $\mathrm{ATT}(d \mid d)$	dose d against untreated units	parallel trends: without treatment, units at dose d would have evolved like the untreated
Slope $\mathrm{ACRT}(d \mid d)$	dose d against nearby doses	**strong** parallel trends: all units would respond alike to the same dose

- The researcher chooses the assumptions depending on the effect they want to analyse.
- In practice, the level is estimated dose bin by dose bin: a 2×2 of each bin against the untreated units (Callaway et al. 2024).
- China shock: every market is exposed to Chinese imports, so there is no untreated group. Only the second row is available: comparisons between doses, under strong parallel trends.

Without strong parallel trends



- Solid curve: the ATT$(d \mid d)$, each point a **different group** of units.
- Its slope adds the causal response and **selection**: the high-dose units respond more.
- Under strong parallel trends, the two curves coincide.

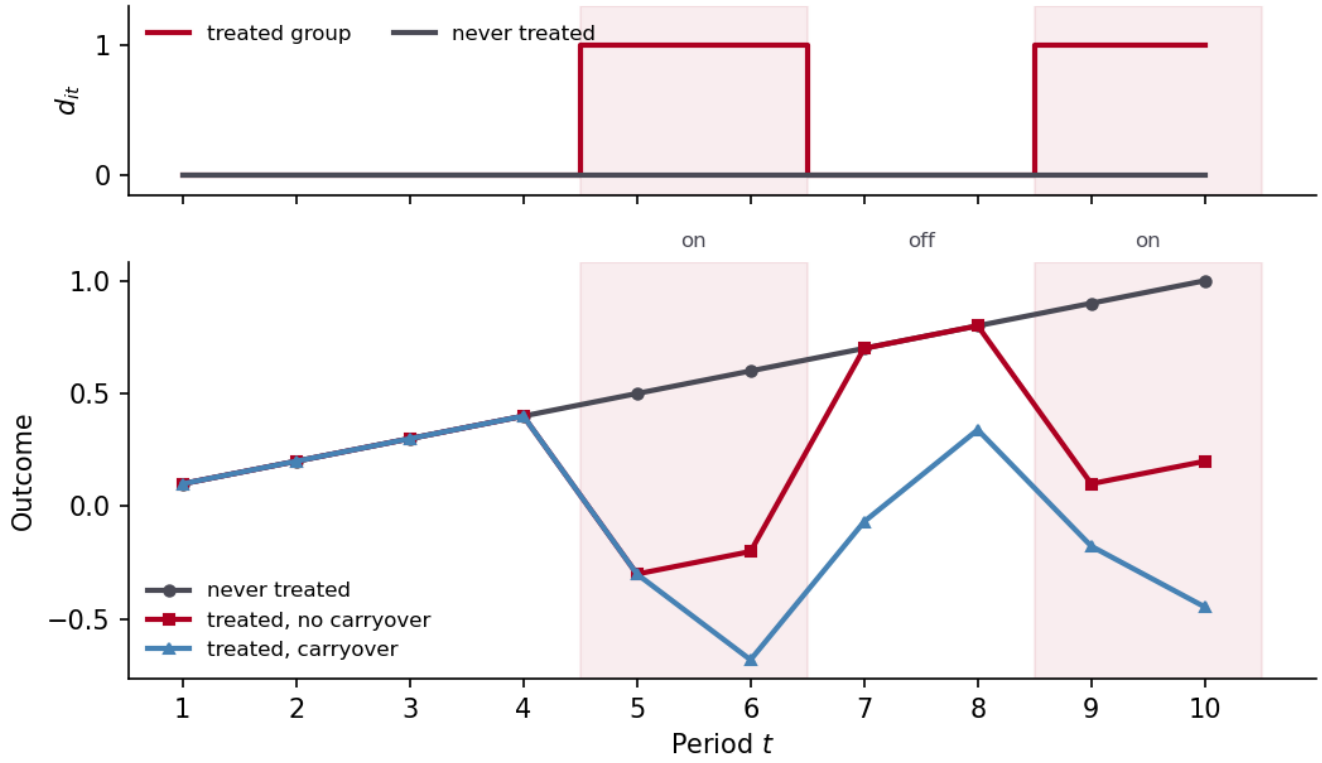
Section 2

On and off

Treatment paths

- When the treatment turns on and off, each unit follows a **path**: treated in some periods, untreated in others.
- Each path is a group: with many periods, there are too many groups to compare one by one.
- Newspapers: a county can gain a newspaper, lose it, and gain another one later.

Carryover



- **Carryover**: past treatment still affects the outcome after it stops (blue line).
- Newspapers: a county that loses its newspaper may keep a higher turnout for some elections.

The switchers event study

- de Chaisemartin and D'Haultfœuille (2024) compare the **switchers**, units whose treatment changes for the first time, with the **not-yet-switchers**, units whose treatment has not changed yet.
- Assumption: without the change, the switchers would have evolved like the not-yet-switchers (parallel trends), and they do not react before the change (no anticipation).
- The effect ℓ periods after the **first change** of treatment: following the actual path, against keeping the initial treatment.
- No restriction on carryover: what happens after the first change is part of the effect.
- Newspapers: the effect on turnout ℓ elections after a county first gains or loses a newspaper.

Reading the switchers event study

- As in the event study of chapter 4, the estimator gives one coefficient per period around the first change:
  - **effects** ($\ell = 1, 2, \ldots$): after the first change;
  - **placebos** ($\ell = -1, -2, \ldots$): the same comparison **before** the first change, close to zero if parallel trends hold. This is the pre-trend test of chapter 4.
- The effect at ℓ includes everything that happened since the first change, switch-offs included. Newspapers: a county that gains a newspaper and loses it two elections later.
- **Normalized** effects divide by the extra treatment received since the first change: the effect of one newspaper, per election.

Section 3

Conclusion

Conclusion

- With a dose, first decide which effect you want: the effect of the dose received needs parallel trends, the effect of one more unit of dose needs strong parallel trends.
- When the treatment turns on and off, compare the units whose treatment changes with the units whose treatment has not changed yet. Each effect is the effect of the whole path since the change.
- In both cases, the TWFE coefficient mixes effects with weights set by the regression: do not report it alone.

Section 4

In R

In R: effects and placebos (DIDmultiplegtDYN)

Four lines estimate the effects and the placebos of the switchers event study.

```
library(data.table); library(fixest); library(DIDmultiplegtDYN)
d <- fread("rica_grippe_td6.csv")[canards_val > 0]                    # 1
d[, y := log(canards_val)]                                           # 2
es <- did_multiplegt_dyn(as.data.frame(d), outcome = "y",
        group = "id", time = "annee", treatment = "restriction",
        effects = 8, placebo = 3, cluster = "id")                    # 3
es$results$p_jointplacebo                                            # 4
```

- **1**: one row per farm and year; only the years with a positive duck output are kept, so that its log exists.
- **2**: the outcome, the log of duck output.
- **3**: switchers against not-yet-switchers: 8 effects after the first change and 3 placebos before, with standard errors clustered by farm. `es$results$Effects` and `es$results$Placebos` give the event-study curve.
- **4**: the joint test that the three placebos are zero.

In R: normalized effects and TWFE

Two lines estimate the normalized effects and the TWFE coefficient to compare.

```
nor <- did_multiplegt_dyn(as.data.frame(d), outcome = "y",
        group = "id", time = "annee", treatment = "restriction",
        effects = 8, placebo = 3, cluster = "id",
        normalized = TRUE)                                    # 5
feols(y ~ restriction | id + annee, d, cluster = ~id)         # 6
```

- **5**: the same estimation with `normalized = TRUE`, in which each effect is divided by the years of treatment accumulated since the first change.
- **6**: the TWFE coefficient, where the years after the switch-off count as untreated.

References I

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Callaway, Brantly, Andrew Goodman-Bacon, and Pedro H. C. Sant'Anna. 2024. *Difference-in-Differences with a Continuous Treatment*. NBER Working Paper No. 32117. National Bureau of Economic Research.

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Gentzkow, Matthew, Jesse M. Shapiro, and Michael Sinkinson. 2011. "The Effect of Newspaper Entry and Exit on Electoral Politics." *American Economic Review* 101 (7): 2980–3018.